What is a Function?

In mathematics, a **function** is a rule or a mapping that takes an input and produces **exactly one output**. It's like a machine: you feed it a value, it does some processing, and spits out the result.

Notation:
A function is usually written as:

$f(x)$

which means: "a function named f, applied to the input x".

Example:
If → $f(x)=x+2$

Then → $f(3)=3+2=5$

Formal Definition:

A function is a relation that assigns **each element of a domain** (input space) to **exactly one element of a range** (output space).

Types of Function Representations:

Functions can be represented in several ways:

- **Algebraic**: Using formulas (e.g. $f(x)=x^2+3x$)
- **Graphical**: As a curve or line on a coordinate system
- **Tabular**: A table listing input-output pairs

Why Functions Matter in AI:

Functions are everywhere in AI:

- **Activation functions** like ReLU and Sigmoid
- **Loss functions** like MSE and Cross-Entropy
- **Gradient Descent** moves "down" the slope of a function to minimize error

Understanding how functions behave—whether they increase, decrease, or have peaks/troughs—is key to training good models.

Bonus Insight:

Any trained machine learning model is essentially a function!

After training, your model becomes a complex mathematical function that maps inputs to outputs.

For example:

- A trained **Linear Regression model** becomes:

  $$f(x)=wx+b$$

- A **Neural Network** becomes a **composition of many functions**, like:

  $$f(x)=\sigma(W3\cdot\sigma(W2\cdot\sigma(W1\cdot x+b1)+b2)+b3)$$

Where σ is an activation function like ReLU or Sigmoid.

So, when you give the model a new input, it simply runs it through this function to give a prediction. That's the whole idea behind inference!

Simple Example:

$f(x)=2x+1 \quad \Rightarrow \quad f(4)=2(4)+1=9$

Summary (Bullet Points):

- A **function** maps input to exactly one output.
- Functions can be algebraic, graphical, or tabular.
- Functions are core to every concept in AI (activation, loss, optimization).
- After training, any ML model becomes a **mathematical function**.
- Neural networks are just **compositions of multiple functions**.